

Linux Users, Groups and File Permissions

Student Study Notes

Users in Linux

A user is an account that can log in to a Linux system. Linux is a multi-user operating system and identifies users through unique IDs (UIDs).

Types of Users

- Root User (UID 0): Complete administrative control.
- Regular Users: Normal users for daily work.
- System Users: Service accounts used by applications such as Apache, MySQL, and Nginx.

User Information

Stored in `/etc/passwd`.

Useful commands:

`whoami`, `who`, `w`, `id`, `useradd`, `userdel`, `usermod`, `passwd`

Groups in Linux

Groups simplify permission management by allowing permissions to be assigned to a collection of users.

Group information is stored in `/etc/group`.

Types of Groups

- Primary Group
- Secondary Groups

Useful commands:

`groupadd`, `groupdel`, `groups`, `usermod -aG`

File Ownership

Every file has:

- Owner
- Group
- Others

Useful commands:

`ls -l`, `chown`, `chgrp`

File Permissions

Permission categories:

- User (u)
- Group (g)
- Others (o)

Permission types:

- Read (r = 4)
- Write (w = 2)
- Execute (x = 1)

Common Permission Values

777 = `rw-rw-rw-rwx`

755 = `rw-r-xr-x`

700 = `rw-x-----`

644 = `rw-r--r--`

600 = rw-----

chmod Examples

```
chmod 777 file.txt
chmod 755 script.sh
chmod 600 secrets.txt
chmod +x script.sh
chmod g+w file.txt
```

Directory Permissions

Read: View filenames
Write: Create/Delete files
Execute: Enter directory using cd

Recursive Operations

```
chmod -R 755 project
chown -R john:developers project
```

Important Commands Summary

whoami, who, w, id, groups, useradd, userdel, usermod, passwd, groupadd, groupdel, ls -l, chmod, chown, chgrp, sudo

Memory Trick

Think of every file as having three locks: Owner, Group and Others.
Each lock can have three keys: Read (4), Write (2) and Execute (1).
Example: chmod 754 file.txt = Owner(rwx), Group(r-x), Others(r--).